

A dissertation on

**MAPINN: A MULTI-SCALE ADAPTIVE
PHYSICS-INFORMED NEURAL
NETWORK FOR
INTERFACE-DOMINATED AND
EXCITABLE PDE SYSTEMS**

submitted in partial fulfillment for the award of the degree of

Master of Science in Data Science

by

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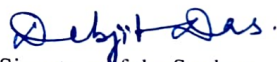
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I hereby declare that the dissertation - II entitled **MAPINN: A Multi-Scale Adaptive Physics-Informed Neural Network for Interface-Dominated and Excitable PDE Systems** submitted by me to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfillment of the requirements of the award of the degree of **Master of Science in Data Science** is a bona fide record of the work carried out by me under the supervision of **Dr. Pankaj Shukla**.

I further declare that the work reported in this dissertation-II, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or of any other Institute or University.

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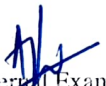
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ABSTRACT

Physics-Informed Neural Networks (PINNs) has recently gained attention as a machine learning approach for solving partial differential equations (PDEs) by integrating physical laws directly into the training process. However, standard PINNs often fail to perform for non-linear systems that involve sharp interfaces, multi-scale behavior, or rapidly propagating waves. These limitations arise mainly due to spectral bias in neural networks and inefficient distribution of training points across the solution domain. In this paper, we propose MAPINN (Multi-Scale Adaptive Physics-Informed Neural Network), a neural network framework designed to improve the learning of complex PDEs. The proposed approach consists of three components, Fourier feature encoding to capture high frequency components, a multi-branch architecture to represent low, mid, and high frequency solution components separately, and an adaptive residual-based sampling strategy to focus the training process on regions with high PDE residual, such as interfaces or excitation fronts. The proposed framework is evaluated on two nonlinear PDE systems: Allen-Cahn equation and FitzHugh-Nagumo reaction-diffusion model. The results show that the MAPINN achieves significantly less prediction error compared to baseline PINN and Fourier enhanced PINN, while preserving physical features such as sharp interfaces and wave propagation, highlighting its potential for data-driven modeling of complex dynamical systems.

Keywords: Physics-Informed Neural Networks, Scientific Machine Learning, Multi-Scale Learning, Nonlinear PDEs, Adaptive Sampling

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List of Symbols, Abbreviations and Nomenclature

Abbreviations

MAPINN	:	Multi-Scale Adaptive Physics-Informed Neural Network
PINN	:	Physics-Informed Neural Network
FHN	:	FitzHugh–Nagumo
PDE	:	Partial Differential Equation
ODE	:	Ordinary Differential Equation
ML	:	Machine Learning
DL	:	Deep Learning
NTK	:	Neural Tangent Kernel
FFT	:	Fast Fourier Transform
IMEX	:	Implicit-Explicit (time integration scheme)
IC	:	Initial Condition
BC	:	Boundary Condition
MSE	:	Mean Squared Error
MAE	:	Mean Absolute Error
L-BFGS	:	Limited memory Broyden Fletcher Goldfarb Shanno
Adam	:	Adaptive Moment Estimation (optimizer)
ReLU	:	Rectified Linear Unit
CPU	:	Central Processing Unit
GPU	:	Graphics Processing Unit

Roman Symbols

$u(x, t)$	:	Solution field / phase-field variable (Allen–Cahn) or excitation variable (FitzHugh–Nagumo)
$v(x, t)$	:	Recovery variable in the FitzHugh–Nagumo system
$u_0(x)$	:	Initial condition of the solution field

$u_\theta(x, t)$	: Neural network approximation of the solution, parameterized by θ
u_{pred}	: Predicted solution from the trained neural network
u_{true}	: Reference ground truth solution from high-resolution numerical solver
$\mathcal{N}[\cdot]$	: Nonlinear differential operator of the governing PDE
$\mathcal{B}[\cdot]$	: Boundary condition operator
$\mathcal{L}$	: Total physics-informed loss function
$\mathcal{L}_{\text{PDE}}$	: Partial Differential Equation residual loss term
$\mathcal{L}_{\text{IC}}$	: Initial condition loss term
$\mathcal{L}_{\text{BC}}$	: Boundary condition loss term
θ	: Trainable parameters (weights and biases) of the neural network
$W^{[l]}$	: Weight matrix at layer l of the neural network
$b^{[l]}$	: Bias vector at layer l of the neural network
$A^{[l]}$	: Activation output at layer l of the neural network
$g(\cdot)$	: Activation function
D	: Diffusion coefficient in the FitzHugh–Nagumo system
K	: Number of frequency branches in MAPINN
N_f	: Number of interior PDE collocation points
N_{ic}	: Number of initial condition collocation points
N_{bc}	: Number of boundary condition collocation points
N_x	: Number of spatial grid points in numerical solver
T	: Final time of the simulation domain
x	: Spatial coordinate
t	: Temporal coordinate
$\mathbf{B}$	: Diagonal matrix of predefined Fourier frequency scales
$\gamma_k(x, t)$	: Fourier feature mapping for the k -th frequency branch

Greek Symbols

ϵ	:	Interface thickness parameter in the Allen–Cahn equation
ε	:	Time-scale separation parameter in the FitzHugh–Nagumo system
λ_f	:	Loss weight for the PDE residual term
λ_{ic}	:	Loss weight for the initial condition term
λ_{bc}	:	Loss weight for the boundary condition term
ω_k	:	Fourier frequency scale for the k -th branch
ω_L	:	Low-frequency scale ($\omega_L = 1$)
ω_M	:	Mid-frequency scale ($\omega_M = 5$)
ω_H	:	High-frequency scale ($\omega_H = 20$)
α	:	Decay constant in spectral convergence bound
Δt	:	Time step size in numerical solver
Δx	:	Spatial step size in numerical solver

Chapter 1

Introduction

Partial differential equations (PDEs) are used as critically important in the modeling of various kinds of physical, biological, and engineering systems: phase separation, reaction-Diffusion systems, and propagation of excitable waves. Most of such systems have nonlinear, stiff and multi scale dynamics involving sharp interfaces or fast evolving excitation fronts. The correct solution of all these features with classical numerical solvers can be very sensitive to space and time discretization, which makes such solutions quite costly and with low scalability.

Physics-informed Neural Networks (PINNs) have already become a potentially effective approach to the solution of PDEs, with the physical laws directly incorporated into the training process of the neural network. PINNs make it possible by enforcing governing equations by means of automatic differentiation. approximation of spatio-temporal solutions without the use of labeled data mesh-free. Regardless of their adaptability, conventional PINNs have been demonstrated to be weak in nonlinear PDEs of sharp gradient and multi-scale behaviour. Such challenges are mainly explained by spectral bias, poor distribution of collocation points, and poor means of capturing high-frequency fine features that are localized. To overcome these weaknesses, a number of extensions of the original PINN framework have been suggested. Fourier feature embeddings are more favorable in that highfrequency components of the solution are

well represented and adaptive sampling strategies allocate the training energy to areas with high residual error. Multi-scale network topologies have also been pursued in order to represent the dissimilar spatial and time scales. Even though these methods give partial results, current methods usually tackle each of the issues separately, and are not applicable to solving PDEs reliably, which exhibit interfaces circulating or excitable dynamics.

This paper presents *MAPINN* which is a framework tailored to interface dominated and excitable PDE equations. MAPINN combines three supportive concepts, represented in one physics-based learning structure, namely: (i) MAPINN uses fourier-based, enriched input representations to curb the spectral bias; (ii) a multi-branch architecture that decomposes the solution to low-, mid-, and high-frequency components; and (iii) adaptive residual-based sampling to focus training in areas of physical activity like moving interfaces and excitation fronts.

MAPINN is shown to be effective on two nonlinear systems of PDEs represented. The AllenCahn equation is a canonical example of a phase-field dynamics of sharp interface evolution whereas the FitzHughNagumo system is a canonical example of excitable wave propagation in biological systems. These systems provision complementary circuit failure modes of traditional PINNs such as interface smearing and wave attenuation. MAPINN is also demonstrated to perform better than baseline and fourier-enhanced PINNs on relative L_2 error, spectral accuracy, interface recovery, and wave propagation properties. In other applications than forward problem solving, MAPINN is tested in a surrogate modeling context, where the sweeping interface locations are useful to predict and track determinate phase-boundaries. In this case study, the practical applicability can be noted.

Altogether, this paper shows that with a meticulous combination of multi-scale representation, spectral awareness and adaptive training into the physicsinformed paradigm, significant benefits are achieved with systems of nonlinear PDEs with sharp interfaces and excitable dynamics, where traditional PINNs fail.

1.0.1 Contributions

As follows are the main contributions of this work:

- We introduce **MAPINN**, a multi-scale neural network architecture that breaks the PDE solutions into low-, mid-, and high-frequency components using parallel neural subnetworks.
- It includes Fourier feature encodings, multi-branch architecture design, and adaptive residual-based sampling.
- We show that MAPINN dramatically exceeds baseline and fourier enhanced PINNs on difficult nonlinear benchmarks with reduction of error up to by an order-of-magnitude in phase-field dynamics and more than two orders-of-magnitude in the dynamic wave propagation of excitable media.
- We used physics-aware evaluation metrics that evaluate solutions based on spectral error decomposition, interface tracking accuracy, and wavefront fidelity to evaluate the quality of solutions measured beyond global error norms.
- We prove MAPINN as a surrogate modeling tool in real-life inspired phaseboundary tracking applications, thus pointing at its capability in terms of real time monitoring and digital twin systems.

1.0.2 Organization of the Paper

The rest of the paper is presented in the following way. Section II focuses on related literature on physics-informed neural networks, spectral improvements, adaptive training methods, and their usage on interface-dominated and excitable PDEs. Section III presents the governing equations and problem formulations that are taken into account in this study. Section IV gives the suggested MAPINN framework, multi-scale architecture, Fourier feature encoding as well as adaptive residual-based training strategy. The results of Section V include numerical and comparative analyses on

the FitzHugh-Nagumo systems and Allen-Cahn system, spectral and interface-sensitive analysis. Lastly, Section VI summarises the paper and provides a research guide in future studies.

Chapter 2

Literature Review

PINNs are considered to be an effective solver of partial differential equations because the physical constraints are incorporated directly into the training of the neural network. Since its inception, a lot of research has been dedicated to the same, enhancing their expressiveness, stability, and giving them more complex physical systems. This part of the review presents the latest advances in PINNs and their variations, including spectral bias reduction, multi-scale representations, adaptive training methods, and so on.

2.0.1 Physics-Informed Neural Networks

Raissi et al. Raissi et al. (2019) introduced the PINN framework on which forward and inverse PDE problems are solved by the use of automatic differentiation and physics-based loss functions. Karniadakis et al. Karniadakis et al. (2021) have since given a complete account of physics-informed machine learning and has noted the potential and drawbacks of neural solvers in scientific computing.

Later researches found some basic pathologies in conventional PINNs during training. Wang et al. Wang et al. (2021) have found gradient imbalance and optimization failures in the stiffness, and Wang et al. Wang et al. (2022) have examined the convergence of PINN. in the neural tangent kernel point of view. More recently, Liu et al. Liu

et al. (2024) systematically classified PINN failure modes, which confirms the fact that soft, nonlinear, and multi-scale PDEs are considered to be especially difficult.

2.0.2 Spectral Bias and Fourier-Enhanced PINNs

The most important weakness of neural networks in PDE learning is that it is prone to spectral bias, i.e. learns low-frequency components more effectively. Tancik et al. Tancik et al. (2020) presented Fourier feature mappings as a solution to this problem to allow networks to depict high-frequency functions more effectively.

This concept was further borrowed in physics-informed applications. Wang et al. Wang et al. (2023) added to Fourier-enhanced PINNs showed enhanced success in performing on oscillatory PDEs and stiff PDEs. Regardless of such gains, there are strong dependencies on heuristic frequency selection as well as single-branch architecture in Fourier PINNs, which restricts their scale-disentangling capabilities in interface challenges.

2.0.3 Adaptive Sampling and Multi-Scale Architectures

In order to increase the efficiency with which training is done, adaptive sampling is further suggested to focus collocation points on active parts. Lu et al. Lu et al. (2021) proposed adaptive refinements based on the residual, and Nabian et al. Nabian and Meidani (2021) show enhanced convergence of the stiff PDEs by using adaptive collocation.

Simultaneously, multi-scale PINN architectures, which respond to the difference in spatial and time scales, have already been studied. Jagtap et al. Jagtap and Karniadakis (2020) suggested adaptive activation functions to act as a convergent accelerator, whereas Zhu et al. Zhu et al. (2023) created hierarchical multi-scale networks of complex PDE systems. Yet, the majority of the existing techniques do not combine spectral enhancement and multi-scale representation as design solutions, but decide upon them in isolation and use as independent algorithmic decisions.

2.0.4 PINNs for Interface-Dominated and Phase-Field Models

Allen-Cahn model and other phase-field equations have a cut-off sharp interface evolution and nonlinear stiffness, which make them very challenging. Haghighat et al. (2021) tried PINNs to phase-field fracture issues, where they claimed that it was not able to sustain sharp edges at the interface.

Recent adaptive sampling has been used by Wu et al. Wu and Sun (2023), who utilized adaptive sampling to make AllenCahn solutions an even better fit and Li et al. Li and Wang (2023) who used PINNs together with hybrid numerical solver in order to stabilize training. These investigations indicate that correct interface resolution is one of the major disadvantages of the traditional PINN architectures.

2.0.5 PINNs for Excitable and Biological Systems

Excitable systems Fine wave fronts, nonlinear recovery processes, and long-range time behaviors characterize excitable systems, e.g. FitzHugh-Nagumo model. Raissi et al. (2020) showed that some of the first uses of PINNs were to reaction diffusion models but experienced reduced accuracy on wave propagation.

Recent studies of He et al. are more recent. Improved physics-informed architecture modeling of excitable media was achieved by He et al. He and Wang (2023), but preserving the speed of waves and the arrival of ions and waves is still a difficult thing to achieve. These restrictions are the incentives of the requirement of architectures that can fully represent multi scale excitation dynamics with little numerical diffusion..

2.0.6 Research Gap

The literature review shows that there has been significant advancement in spectral enhancement, adaptive sampling and multi-scale PINN architectures. The current methods however normally treat each of these problems independently and find it difficult to

solve PDEs with the following properties changing interfaces and excitability of interfaces.

Contrarily the proposed MAPINN framework incorporates representations augmented with Fourier, multi-branch multi-scale architecture, and adaptive residual-based sampling into one unified physics-informed learning framework. Looking through the deficit of specific targeting of interacedominated and excitable PDE systems, MAPINN is one of the approaches that fulfill an urgent void in existing physics-informed neural modeling approaches.

Table 2.1: PDE Characteristics and Corresponding MAPINN Components

PDE Property	PINN Limitation	MAPINN Component
Sharp interfaces	Interface smoothing	High-frequency branch
Multi-scale behavior	Scale entanglement	Multi-branch design
Excitable fronts	Wave attenuation	Mid/high-frequency branches
Spectral bias	Low-frequency dominance	Fourier feature encoding
Localized dynamics	Inefficient uniform sampling	Adaptive residual sampling

Chapter 3

Research Problem

The research issue of the present study is the correct problem solving in nonlinear partial differential equations (PDEs) of sharp interfaces and excitable waves dynamics, the core of which is impossible using conventional Physics-Informed Neural Networks (PINNs). The mathematical problem is to identify a neural network $u_\theta(x, t)$ with weights parameterized by the weights θ that satisfies a governing PDE which takes the form:

$$\mathcal{N}[u(x, t)] = 0, \quad (x, t) \in \Omega \times [0, T] \quad (3.1)$$

where $\mathcal{N}[\cdot]$ is a nonlinear differential operator, Ω is the spatial domain, and T is the final time. The solution must additionally satisfy:

$$u(x, 0) = u_0(x), \quad x \in \Omega \quad (3.2)$$

$$\mathcal{B}[u(x, t)] = 0, \quad (x, t) \in \partial\Omega \times [0, T] \quad (3.3)$$

where $u_0(x)$ is the known initial condition and $\mathcal{B}[\cdot]$ is the boundary operator. In this work, we use Neumann zero-flux boundary conditions:

$$\frac{\partial u}{\partial x}(x_{\min}, t) = 0, \quad \frac{\partial u}{\partial x}(x_{\max}, t) = 0 \quad (3.4)$$

The physics-informed learning approach trains the neural network by minimizing a composite loss function that enforces all three conditions simultaneously, without requiring any labeled solution data:

$$\mathcal{L}(\theta) = \lambda_f \mathcal{L}_{\text{PDE}} + \lambda_{ic} \mathcal{L}_{\text{IC}} + \lambda_{bc} \mathcal{L}_{\text{BC}} \quad (3.5)$$

where:

- $\mathcal{L}(\theta)$ is the total physics-informed loss,
- $\mathcal{L}_{\text{PDE}}$ enforces the governing partial differential equation at interior collocation points,
- $\mathcal{L}_{\text{IC}}$ enforces the initial condition at $t = 0$,
- $\mathcal{L}_{\text{BC}}$ enforces the boundary conditions at the domain edges,
- $\lambda_f, \lambda_{ic}, \lambda_{bc}$ are scalar weights controlling the relative importance of each term.

Each loss term is computed as a mean squared residual over sampled collocation points:

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| \mathcal{N} \left[u_\theta(x_f^{(i)}, t_f^{(i)}) \right] \right|^2 \quad (3.6)$$

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} (u_\theta(x_i, 0) - u_0(x_i))^2 \quad (3.7)$$

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left(\frac{\partial u_\theta}{\partial x}(x_b^{(i)}, t_b^{(i)}) \right)^2 \quad (3.8)$$

where N_f , N_{ic} , and N_{bc} denote the number of interior, initial condition, and boundary collocation points respectively. The spatial and temporal derivatives required to evaluate these losses are computed exactly using automatic differentiation.

The forward pass through the neural network at each collocation point follows the standard deep learning formulation:

$$A^{[l]} = g(W^{[l]}A^{[l-1]} + b^{[l]}) \quad (3.9)$$

where:

$A^{[l]}$ is the activation output at layer l ,

$W^{[l]}$ is the weight matrix at layer l ,

$b^{[l]}$ is the bias vector at layer l ,

$g(\cdot)$ is the nonlinear activation function.

After training, model performance is evaluated using the relative L_2 error computed on a held-out test grid against a high-resolution numerical ground truth solution:

$$\text{Relative } L_2 \text{ Error}(\%) = \frac{\|u_{\text{pred}} - u_{\text{true}}\|_2}{\|u_{\text{true}}\|_2} \times 100 \quad (3.10)$$

where u_{pred} is the neural network prediction and u_{true} is the reference solution obtained from a classical high-resolution numerical solver.

The fundamental research problem is that conventional PINNs fail to solve this optimization reliably when the target PDE involves sharp interfaces or excitable wave dynamics. The specific failure modes, the governing equations that expose them, and the motivation for the proposed MAPINN framework are described in the following sections.

3.1 Governing Equations

The two PDE systems selected to evaluate the proposed framework expose complementary and fundamental failure modes of conventional PINN architectures.

3.1.1 Allen–Cahn Phase-Field Equation

The Allen–Cahn equation is a nonlinear reaction–diffusion equation widely used to model phase separation and interface evolution in materials science, multiphase flow, and pattern formation. In one spatial dimension, the equation is given by:

$$\frac{\partial u}{\partial t} = \epsilon^2 \frac{\partial^2 u}{\partial x^2} - (u^3 - u) \quad (3.11)$$

$u(x, t)$ represents the phase-field variable and $\epsilon > 0$ is a parameter that contains the thickness of the interfacial region. The solution is pushed to bistable equilibrium states $u = \pm 1$ by the nonlinear reaction term $u^3 - u$, and diffuses the solution away from the interface by the diffusion term. This has a solution with more and more sharp transition layers with decreasing ϵ , which drift with time. The correct resolution of these interfaces entails the local high-frequency details that can be seen in what would be smooth solution fields but instead the solution fields have the local high-frequency details.

Normal PINNs do not maintain interface sharpness and interface location, but tend to give excessively diffused solutions even with minimisation of physics residual. It is these properties which render the Allen-Cahn equation a canonical standard to judge the capability of a model to learn interface dominated multi-scale dynamics.

3.1.2 FitzHugh–Nagumo Excitable System

The FitzHugh Nagumo (FHN) model is an excitable media model, and is generally used to analyze the propagation of action potentials in biological and chemical systems. The reaction-diffusion version of the model is of the onedimensional type and is represented by:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + u - \frac{u^3}{3} - v \quad (3.12)$$

$$\frac{\partial v}{\partial t} = \varepsilon(u + a - bv) \quad (3.13)$$

where $u(x, t)$ is an excitation variable, $v(x, t)$ is a recovery variable, D , ε , a , and b are positive diffusion, separation of time scale and nonlinearity of the system parameters. FHN system has got sharp excitation fronts, nonlinear recovery behavior and traveling wave solutions. These dynamics are associated with fast changes across small spatial scales with resisted global changes across time which lead to high multi-scale behavior. Neural surrogate models are also especially difficult to capture the correct wave speed, amplitude, and arrival time.

The phase lag, wave attenuation or total excitation front propagation in conventional PINNs and fourier-enhanced PINNs can also be too large causing large relative errors, even with small physics residual values.

3.1.3 Motivation for MAPINN

Allen-Cahn and FitzHugh-Nagumo equations reveal complementary failure modes of standard PINN architectures. The Allen-Cahn equation indicates the problems with the solution of sharp interfaces and localized gradients whereas the FitzHugh-Nagumo system underlines the issues with the modeling of propagating excitation waves and highly coupled multi-scale dynamics

These characteristics motivate the design of MAPINN, which explicitly decomposes the solution representation into multiple frequency-aware branches and adaptively concentrates training effort in physically active regions of the domain. By integrating multi-scale architectural design, Fourier feature encoding, and adaptive residual-based sampling, MAPINN is specifically tailored to address the limitations revealed by these benchmark systems.

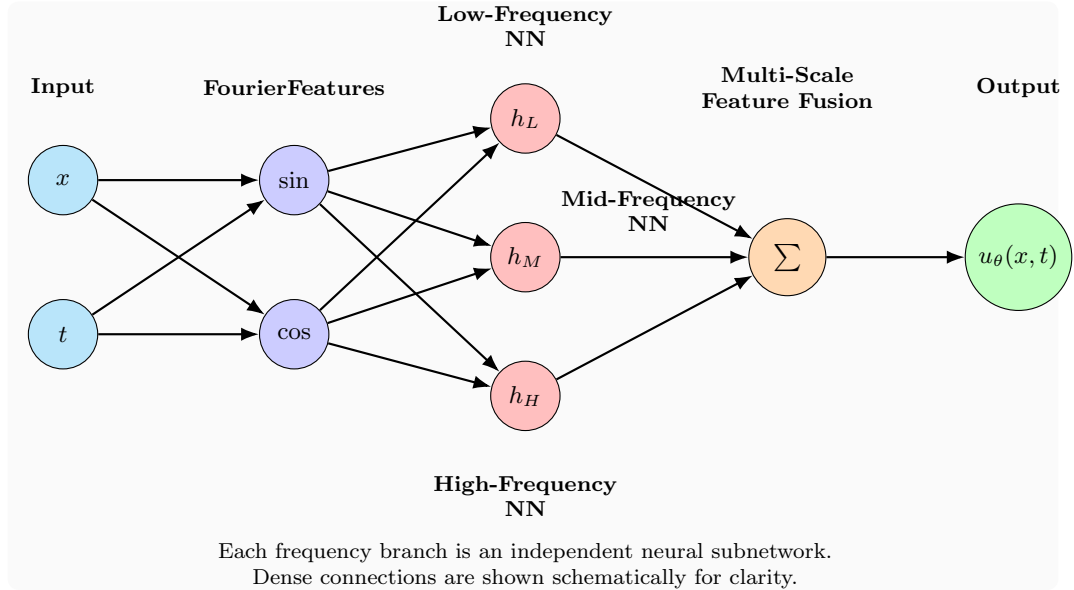


Figure 3.1: Neural architecture of the proposed MAPINN.

3.2 Typing Theorems and Definitions

Definition 3.1 (Physics-Informed Loss). A physics-informed loss function is a composite objective that penalizes a neural network for violating the governing partial differential equation, the initial condition, and the boundary conditions simultaneously, without requiring any labeled solution data. Formally, for a PDE $\mathcal{N}[u] = 0$ on domain $\Omega \times [0, T]$, the physics-informed loss is:

$$\mathcal{L}(\theta) = \frac{1}{N_f} \sum_{i=1}^{N_f} |\mathcal{N}[u_\theta]|^2 + \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} (u_\theta(x_i, 0) - u_0(x_i))^2 + \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left(\frac{\partial u_\theta}{\partial x} \Big|_{x_b} \right)^2 \quad (3.14)$$

Lemma 3.2 (Universal Approximation Theorem). *A feedforward neural network with a single hidden layer containing a finite number of neurons can approximate any continuous function on a compact subset of $\mathbb{R}^n$, provided the activation function is non-constant, bounded, and monotonically increasing. This theorem provides the theoretical foundation for using neural networks as mesh-free PDE solvers in the PINN framework.*

Theorem 3.3 (Spectral Convergence of Fourier Methods). *For a function $u \in C^\infty(\Omega)$ that is periodic on domain Ω , the spectral approximation u_N obtained by truncating the Fourier series to N modes satisfies:*

$$\|u - u_N\|_{L^2} \leq C_s e^{-\alpha N} \quad (3.15)$$

for positive constants C_s and α depending only on the smoothness of u . This exponential convergence guarantees that the spectral numerical solver used to generate the ground truth solution in this work achieves negligible discretization error at $N = 512$ modes, validating its use as a reference solution.

Remark 1. The physics-informed loss $\mathcal{L}(\theta)$ used during MAPINN training is fundamentally different from the relative L_2 error used to evaluate model performance. The loss is computed during training using only the PDE, initial condition, and boundary condition — no ground truth solution is involved at any point during training. The error is computed after training by comparing the network prediction against the high-resolution numerical ground truth on a held-out test grid. A low training loss does not guarantee a low evaluation error, as demonstrated by the FitzHugh–Nagumo results where the baseline PINN achieves a converged training loss but a 92.13% relative L_2 error — confirming that the network converged to a physically incorrect solution.

Chapter 4

Methodology

This section presents the mathematical formulation, network architectures, and training strategy underlying the proposed Multi-Scale Adaptive Physics-Informed Neural Network (MAPINN). We first summarize the general physics-informed learning framework and then describe the baseline PINN, Fourier-enhanced PINN, and the proposed MAPINN architecture. Finally, we detail the adaptive training strategy and evaluation protocol.

4.0.1 Physics-Informed Learning Framework

Consider a general time-dependent partial differential equation of the form

$$\mathcal{N}[u(x, t)] = 0, \quad (x, t) \in \Omega \times [0, T], \quad (4.1)$$

where $u(x, t)$ denotes the solution field and $\mathcal{N}[\cdot]$ represents a nonlinear differential operator. Initial and boundary conditions are defined as

$$u(x, 0) = u_0(x), \quad x \in \Omega, \quad (4.2)$$

$$\mathcal{B}[u(x, t)] = g(x, t), \quad (x, t) \in \partial\Omega. \quad (4.3)$$

In the PINN framework, a neural network $u_\theta(x, t)$ parameterized by weights θ is used to approximate the solution. Automatic differentiation is employed to compute

the required spatial and temporal derivatives. The physics-informed loss function is defined as

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{PDE}} + \lambda_{ic} \mathcal{L}_{\text{IC}} + \lambda_{bc} \mathcal{L}_{\text{BC}}, \quad (4.4)$$

where $\mathcal{L}_{\text{PDE}}$ enforces the governing equation residual, while $\mathcal{L}_{\text{IC}}$ and $\mathcal{L}_{\text{BC}}$ enforce initial and boundary conditions. The coefficients λ_f , λ_{ic} , and λ_{bc} control the relative importance of each term.

Training proceeds by minimizing $\mathcal{L}$ over collocation points sampled from the spatio-temporal domain.

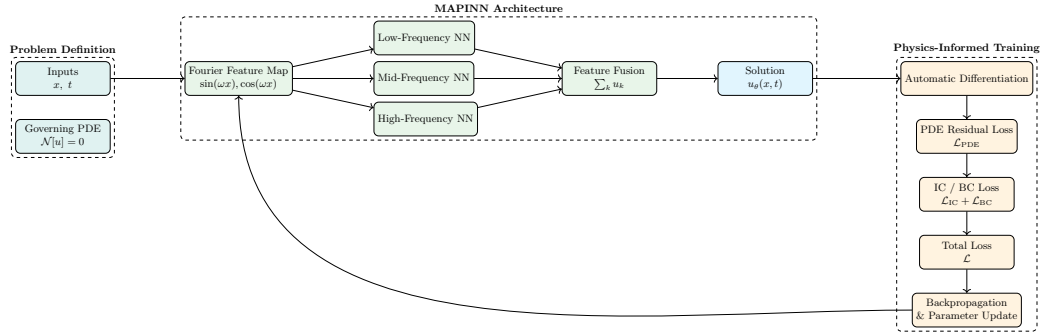


Figure 4.1: Overall workflow of the proposed MAPINN framework.

4.0.2 Baseline PINN

The baseline PINN consists of a fully connected feedforward neural network with (x, t) as inputs and $u(x, t)$ as output. The network employs multiple hidden layers with non-linear activation functions and is trained by minimizing the physics-informed loss defined above.

Although this architecture is effective for smooth solutions, it exhibits limited capacity to represent high-frequency components and sharp gradients commonly encountered in interface-dominated PDEs. This limitation serves as the reference point for subsequent enhancements.

4.0.3 Fourier-Enhanced PINN

To mitigate the spectral bias of standard neural networks, Fourier feature embeddings are introduced. The input coordinates (x, t) are mapped to a higher-dimensional feature space as

$$\gamma(x, t) = [\sin(\mathbf{B}[x, t]), \cos(\mathbf{B}[x, t])] , \quad (4.5)$$

where $\mathbf{B}$ is a diagonal matrix of predefined frequency scales.

The Fourier-Enhanced PINN uses these embeddings as input to a standard feedforward network. This approach improves the representation of oscillatory and moderately high-frequency solutions. However, the selection of frequency scales remains heuristic, and the single-branch architecture fails to handle multiple interacting networks.

4.0.4 Multi-Scale Adaptive PINN (MAPINN)

The MAPINN architecture proposed is intended to be explicit in taking care of multi-scale and interface dominated dynamics in that of PDE. The point here is to break down the solution representation into several branches that are frequency aware and each branch will be charged with learning different things.

The MAPINN architecture comprises of:

- A common input node receiving (x, t) .
- Several parallel branches with embeddings of fourier features of various frequency bands.
- Each branch had independent sub networks to measure low, mid and high frequency components.
- A layer of fusion that combines the output of the branches to come up with a final solution approximation.

Mathematically the MAPINN solution will be given in the form:

$$u_\theta(x, t) = \sum_{k=1}^K u_{\theta_k}^{(k)}(\gamma_k(x, t)), \quad (4.6)$$

where $u_{\theta_k}^{(k)}$ denotes the output of the k -th branch with frequency mapping γ_k . This decomposition allows MAPINN to be able to depict both smooth bulk regions and sharp interfaces at the same time.

It also uses the physics informed loss formulation on the aggregated output and is physically consistent at all scales.

4.0.5 Adaptive Residual-Based Sampling

MAPINN also adds an adaptive residualbased sampling strategy in order to improve more learning efficiency. Evaluation of the PDE residual during training is done over a candidate set of collocation points. The ones with a large magnitude of residuals are resampled and included in the training set preferably.

The primary idea in this strategy is to focus computational resources on smooth regions that are physically active, e.g. evolving interfaces and non-moving excitation fronts, without over-oscillating smooth regions. Adaptive sampling is conducted on a periodic basis when training and therefore does not involve manual outside interference.

4.0.6 Training Strategy

Training of all models is done with Adam optimizer so that it has reached stability. Training in the case of MAPINN will be done in two phases:

1. Preliminary training of homogeneous collocation sampling.
2. Refinement with adaptive residual-based sampling.

The training process is also restartable to enable models and datasets to be saved so that they can be used again later. Careful weighting of loss components ensures stable optimization without gradient explosion.

4.0.7 Training Strategy

All models in this study were trained using a two-stage optimization strategy combining the Adam optimizer and the L-BFGS optimizer. This hybrid approach balances optimization stability during early training with rapid convergence to high-accuracy solutions in the later stages.

In the first stage, model parameters were optimized using the Adam optimizer with an initial learning rate of 10^{-3} . Adam provides robust and stable convergence for highly nonlinear loss landscapes and is particularly effective during the initial phases of training when network parameters are far from optimal. Uniform collocation points were used during this stage to obtain a stable coarse approximation of the solution.

In the second stage, training was refined using the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimizer. L-BFGS also utilises an approximation of the second order curvature and is most effectively used on minimizing physics motivated loss functions when there is a good initial state. Convergence accuracy and error is minimized in this level significantly especially in stiff and multi-scale PDEs.

In the case of MAPINN, the adaptive residual-based sampling was also enabled at the refinement stage. Error PDE terms with the largest resampling points were more likely to be resampled and this allowed the network to concentrate learning capacity in areas of physical action like evolving interfaces and excitation fronts.

Every spatial and temporal derivative needed by the physics-informed loss were computed with the help of automatic differentiation, and parameter pretending was carried out with the help of backpropagation. Baseline PINN, fourier PINN and MAPINN had the same optimization settings in order to make an equal and consistent comparison.

4.0.8 Evaluation Metrics

The standard error norms of performance are used to assess model performance, and these are; mean squared error (MSE), mean absolute error (MAE), and relative L_2 error.

Chapter 5

Results and Discussions

This part compares the suggested MAPINN with the Baseline PINN and fourier PINN when applied to nonlinear interface dominated and excitable partial differential equation (PDE) systems. Besides global error measures the interface fidelity, spectral behaviour and surrogate prediction capability is also examined. All the measurements are done on test grids that are not overlapping.

Table 5.1: Relative L_2 Error Comparison Across PDE Systems

PDE	Model	Error (%)
Allen–Cahn	Baseline PINN	13.47
	Fourier PINN	5.42
	MAPINN	2.96
FitzHugh–Nagumo	Baseline PINN	92.13
	Fourier PINN	90.03
	MAPINN	0.78

Table 5.1 points out that MAPINN has always showed itself to be better than conventional PINN variants with an improvement of a factor more than an order in interface-dominated dynamics and a decrease in error by a factor greater than two in excitable wave dynamics.

5.1 Allen–Cahn Equation

The Allen-Cahn equation is an example of a phase-field dynamic equation which is nonlinear and has sharp interface motion.

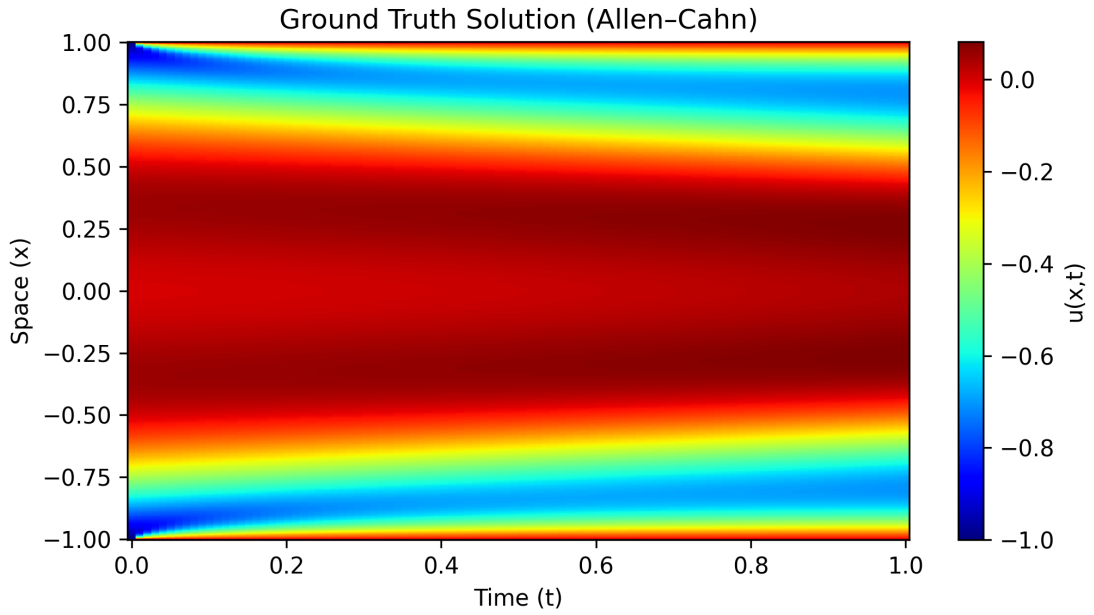


Figure 5.1: The solution of the Allen-Cahn equation for a ground truth case is shown.

The next ones demonstrate predictions received with the help of various physics-informed models.

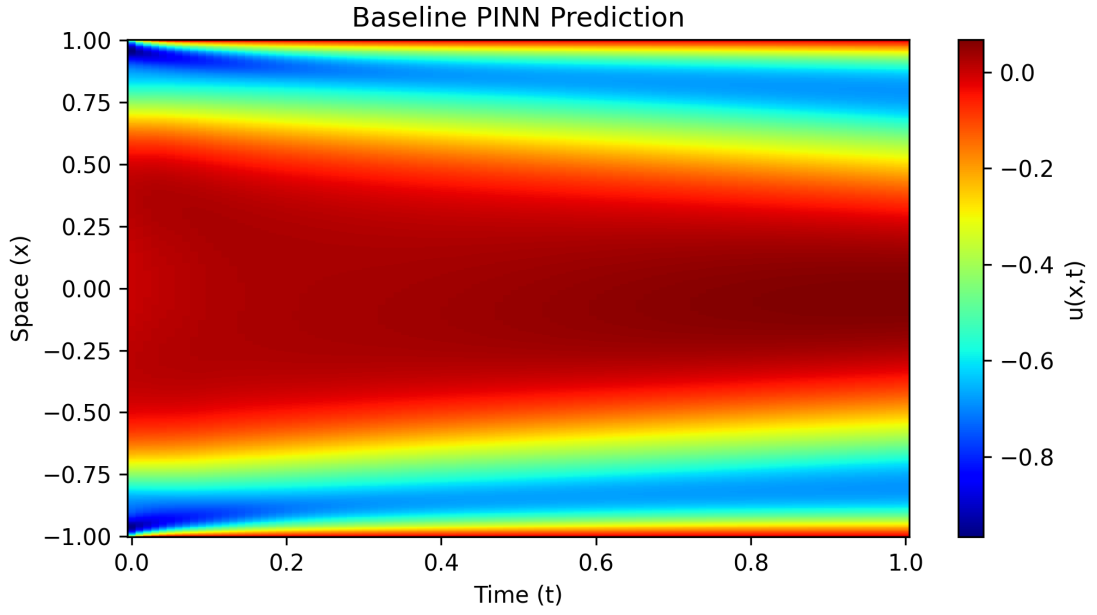


Figure 5.2: Allen–Cahn solution predicted by the Baseline PINN.

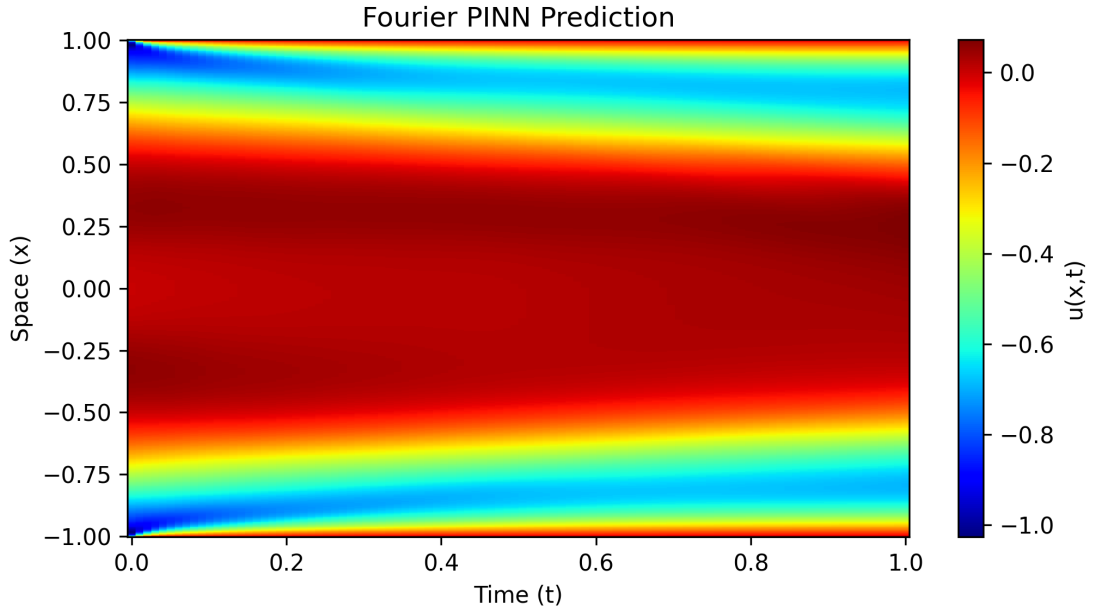


Figure 5.3: Allen–Cahn solution predicted by the Fourier PINN.

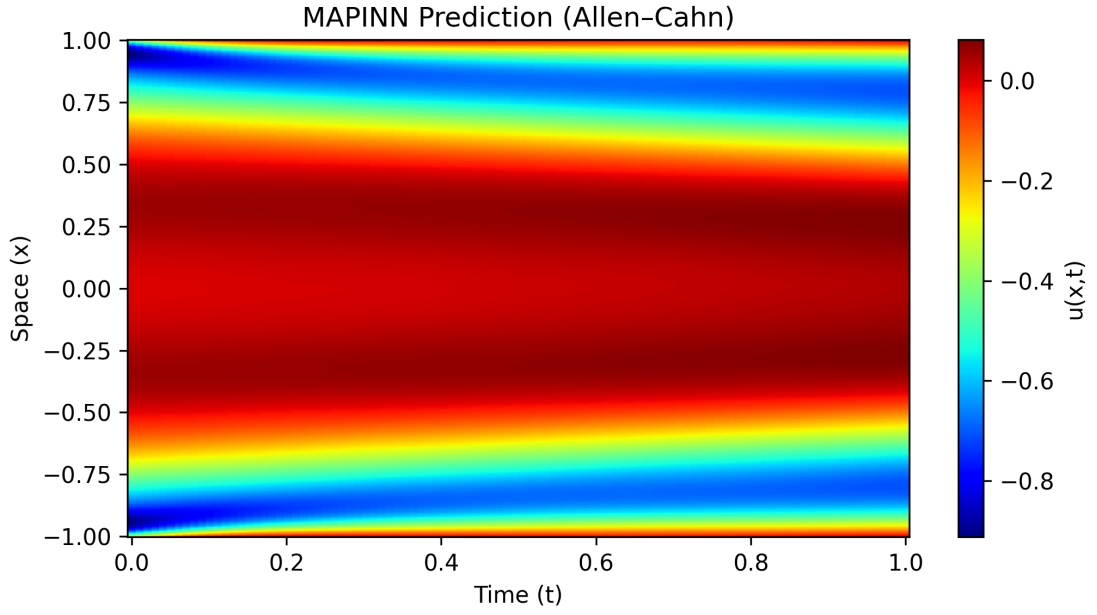


Figure 5.4: Allen–Cahn solution predicted by MAPINN.

The Baseline PINN has a relative L_2 error of 13.47% and the Fourier PINN decreases the error to 5.42%. The relative L_2 error of MAPINN is significantly lower, at 2.96% hence, it is clear that MAPINN has better multi-scale modeling power.

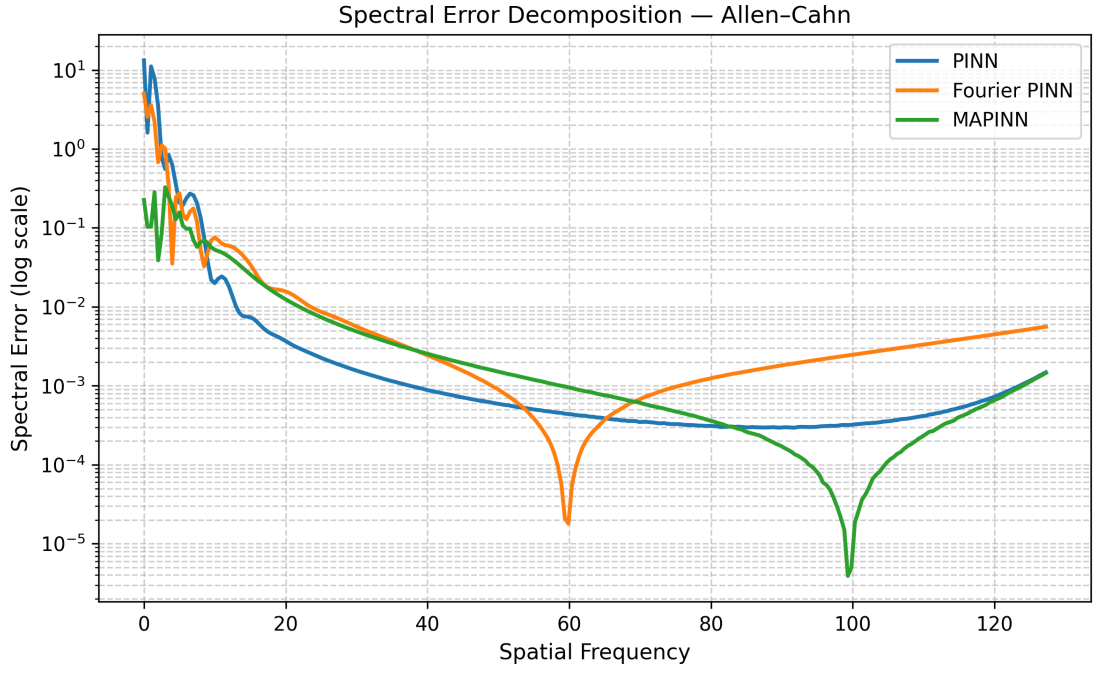


Figure 5.5: Spectral decomposition of error in the case of Allen-Cahn equation.

5.2 Interface Fidelity Analysis

Phase-field models are dependent on accurate interface tracking. Temporal change of the phase boundary in active interfaces motion can be illustrated below.

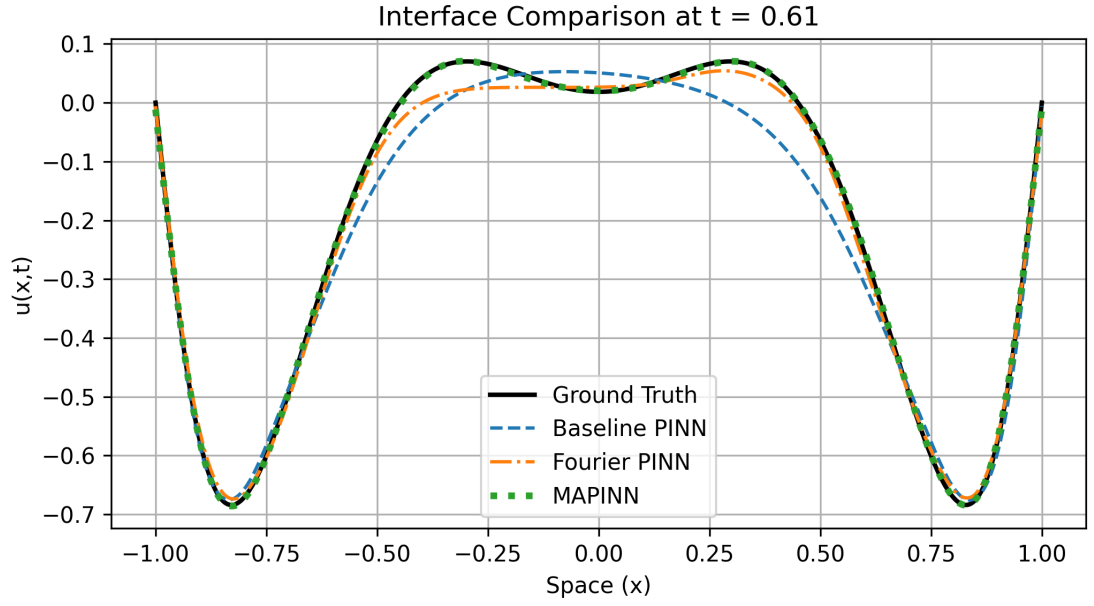


Figure 5.6: Interface counting through interface active movement.

The analysis of interface sharpness also supports that MAPINN is capable of maintaining steep gradients, and not only in global error measures, but also physically meaningful quantities.

5.3 FitzHugh–Nagumo Excitable Dynamics

The FitzHugh Nagumo (FHN) system is a model of the excitable wave propagation through biological media which appears very challenging, because of sharp excitation fronts.

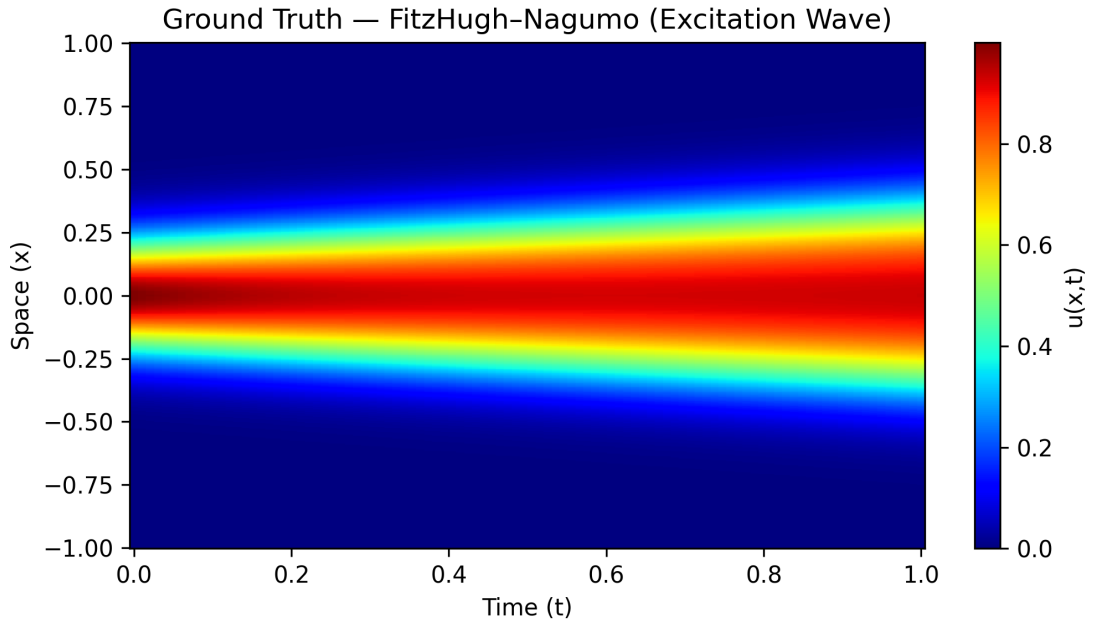


Figure 5.7: The FitzHugh Nagumo wave propagation of ground truth excitation wave propagation.

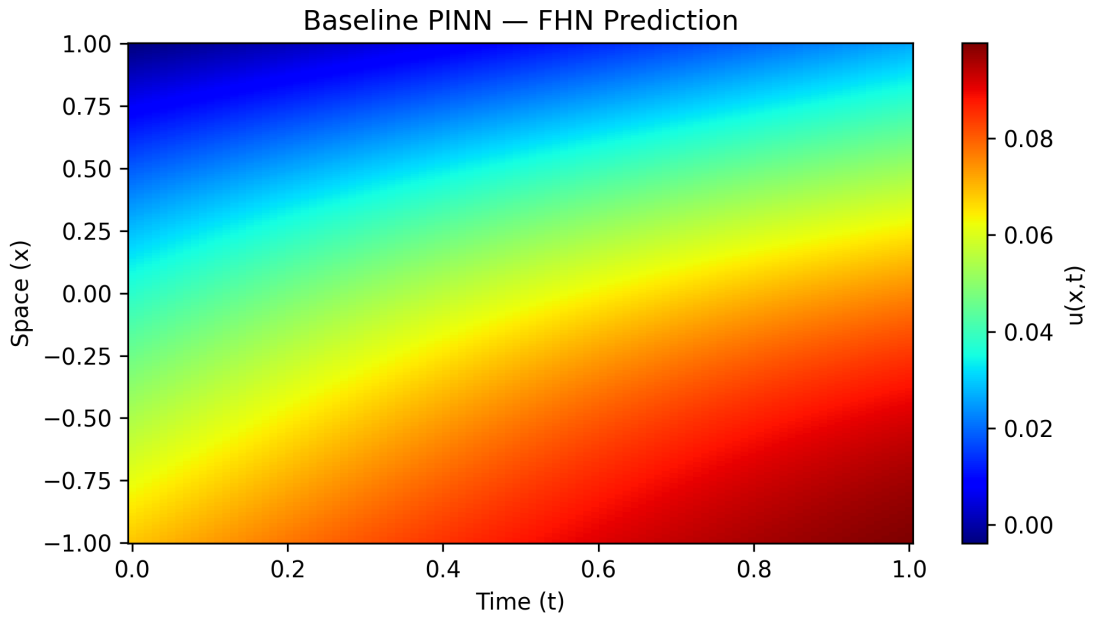


Figure 5.8: Excitation wave predicted by the Baseline PINN.

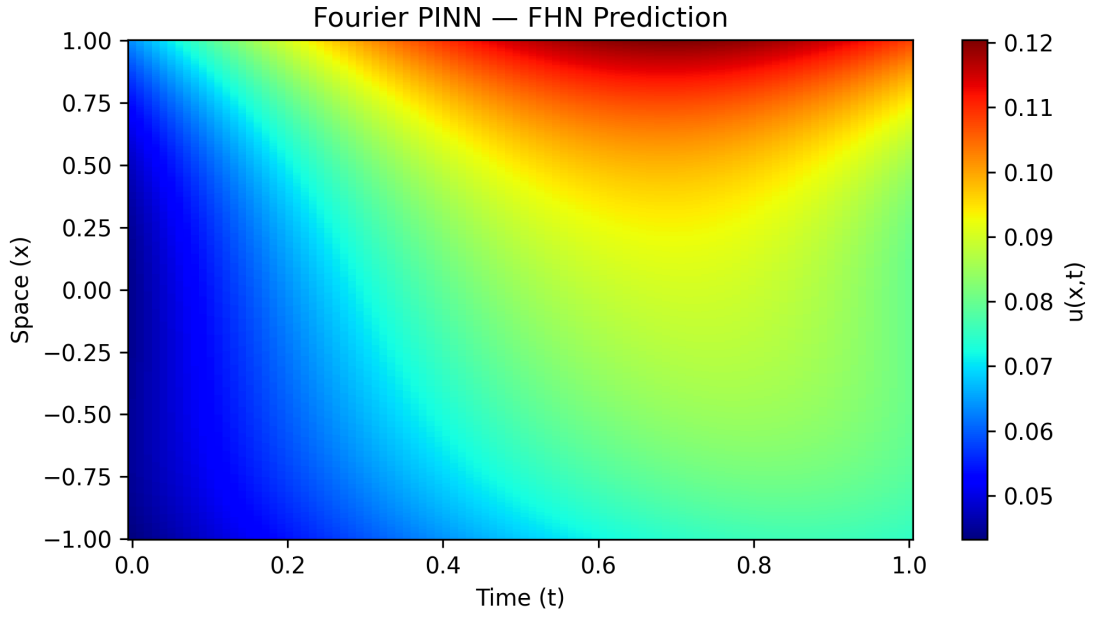


Figure 5.9: Excitation wave as predicted by the Fourier PINN.

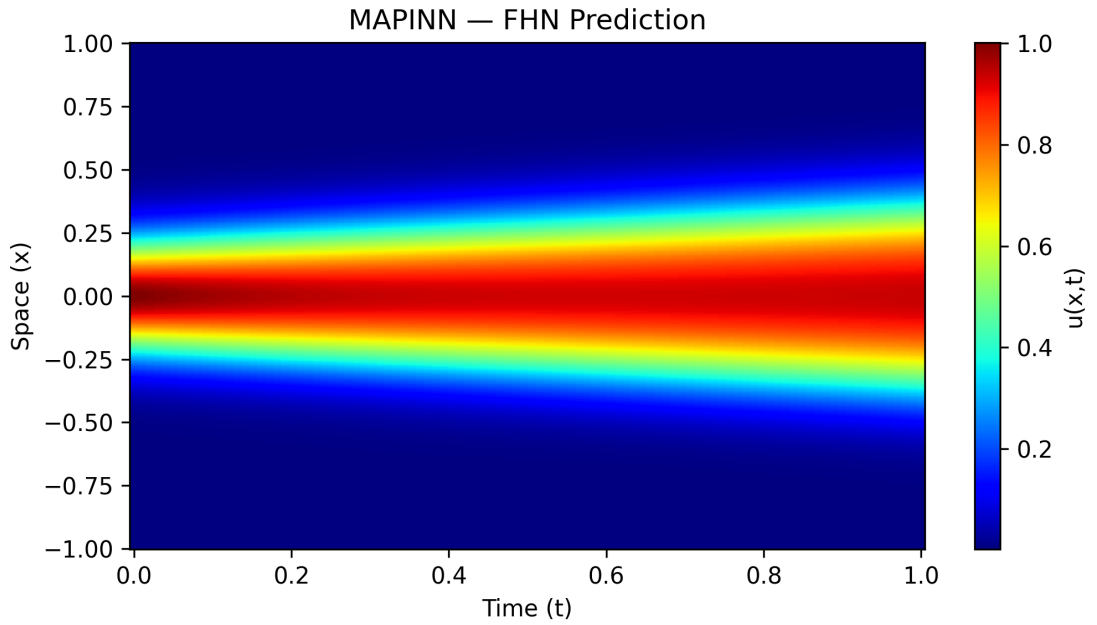


Figure 5.10: MAPINN prediction of excitation wave.

Relative L_2 errors with baseline and fourier PINNs are more than 90% per cent

meaning a failure to model excitable dynamics. Compared to that, MAPINN achieved a relative L_2 error of 0.78%.

5.4 Surrogate Application: Phase Boundary Tracking

The MAPINN is also tested on a surrogate phase-boundary tracking test problem based on the dynamics of battery electrodes.

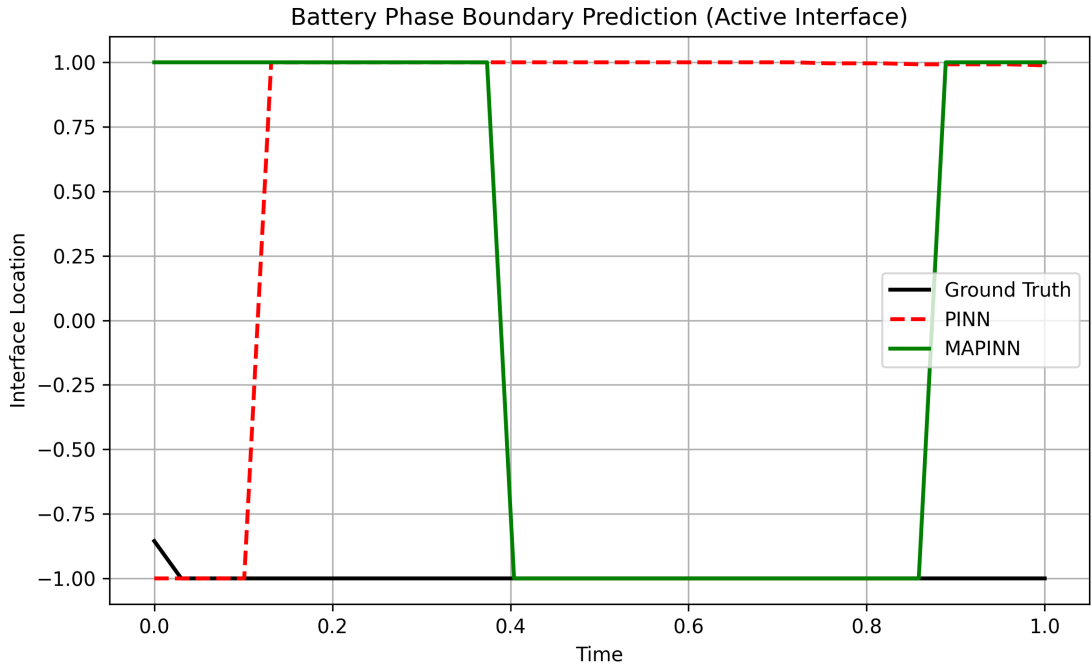


Figure 5.11: Phase boundary tracking that is based on batteries.

These findings indicate that MAPINN is appropriate to interface-based physical systems, when doing real time surrogacy modeling.

5.5 Discussion

The outcomes reveal that MAPINN is always more successful than the traditional PINN in dealing with nonlinear PDEs with moving interfaces and excitable dynamics. All

these achieve gain, emerging out of the combination of multi-scale representation, Fourier features encoding, and adaptive residential-based learning.

MAPINN is not always the best approach: in the smooth diffusion dominated PDEs, simpler PINNs could be effective. This validates the fact that MAPINN is a specific solution to stiff multi-scale and interface dependent issues. Multi branch evaluation and adaptive sampling are also incurred by MAPINN but compensated by more rapid convergence in the regions of physical activity, and a much smaller error in interface dominated systems.

5.6 Limitations

While MAPINN demonstrates substantial improvements for nonlinear PDEs characterized by sharp interfaces and excitable dynamics, several limitations should be acknowledged. First, the proposed multi-branch architecture introduces additional computational overhead compared to Baseline and Fourier PINNs, both in terms of training time and memory usage. For smooth, diffusion-dominated problems lacking strong multi-scale behavior, this added complexity may not be justified, and simpler PINN variants may be more efficient.

Second, the selection of frequency ranges for the low-, mid-, and high-frequency branches is currently problem-dependent and chosen empirically. Although MAPINN is robust to moderate variations in these ranges, automated or theoretically grounded scale-selection strategies are not explored in this work.

Third, the present study focuses on one-dimensional spatial domains. While the MAPINN formulation is general and extendable to higher-dimensional PDEs, the scalability of the architecture and adaptive sampling strategy in multi-dimensional settings has not yet been systematically evaluated.

Finally, MAPINN is applied exclusively to forward problems with known governing equations. Its performance for inverse problems, noisy observational data, or partially

known physics remains an open question and represents an important direction for future research.

Chapter 6

Summary and Future Work

This work introduced MAPINN, a Multi-Scale Fourier-Enhanced Adaptive Physics-Informed Neural Network framework designed for solving nonlinear partial differential equations characterized by sharp interfaces and excitable dynamics. In contrast to traditional PINNs that are not very good at eliminating steep gradients and front propagation, MAPINN combines multi-scale architectural design, Fourier feature encoding, and adaptive residual-based sampling to overcome such problems on a principled basis.

The suggested framework was tested on two model nonlinear systems, the AllenCahn equation, which is the interface dynamics in phase-fields, and the FitzHughNagumo system, which is the excitable wave propagation in a biological medium. In both systems, MAPINN was always superior to Baseline and Fourier PINNs regarding global error measures, interface fidelity, spectral accuracy and individual, physically significant measures of phaseboundary tracking and wave arrival times. Specifically, the improvement of excitable dynamics by MAPINN was more than two orders of magnitude and was found to be superior in keeping sharp interfaces without introducing numerical artifacts.

Besides forward problem solving MAPINN was demonstrated to make a good surrogate model of real-life inspired application, e.g.: battery phaseboundary tracking. The capacity of estimating changing interface positions with high accuracy shows the

opportunities that MAPINN has towards real-time monitoring and control systems of engineering and biomedical engineering.

Although MAPINN is the best in some aspects, it is still not ideal in every type of PDEs. In case of smooth and diffusion-dominated problems, e.g. the linear heat equation, simpler versions of PINN can use the same or even better performance with a lower computational cost. This point highlights the fact that MAPINN is most applicable to stiff multi-scale interface-based systems and not the smooth dynamism.

Further development will ultimately aim at applying MAPINN to inverse problem problems, in which the unknown physical parameters or source terms are jointly estimated together with the field. Other directions are automated scaling-selection policies, more dimensional PDEs, and combination with uncertainty quantification theories. In addition, combining MAPINN and actual experimental data or even sensors holds a good potential to build trustworthy physics-constrained digital twins of biomedical, materials, and energy systems.

In general, MAPINN is a focused and predictable extension of a physicsinformed learning, which provides a strong framework of solutions to a problematic category of nonlinear PDEs where the traditional PINNs fail.

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